DS	**Session**	Hashing	**Question Information**	Level 1 Challenge 92

Problem Description:

Canthi and Sami are having a game! The game is extremely similar to chess, but there is only one piece on the board, which is the Queen. In addition, Queen may only go to the top left corner.

For clarification, If Queen is placed at i,j then in a turn queen can move:
1) Any number of cells leftwards.
2) Any number of cells upwards.
3) Any number of cells Diagonally(only N-W direction).

Please note that board is quarter infinite i.e there is a top left corner but it extends to infinity in south and east direction..

Functional Description

1) Canthi will always play the first turn.
2) They both will get alternative turns.
3) They must move the queen in their turn (No skip option) and that too according to the rule specified above.
4) Whosoever is unable to make a move loses.

Given The position of queen on the board(0 index based). Print who will win the game.

Constraints:
1<=t<=10000
0<=a,b<=1000000

Input:
First line of input contains an integer t - no of test cases.
Each test case is described as a line containing two integers a and b. which is the position of queen on the board.

Output:
print the name of person who will win the game.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
1 2
2 3
```

EXPECTED OUTPUT

```
sami
canthi
```

Test Case 2

INPUT (STDIN)

```
3
5 6
1 5
2 3
```

EXPECTED OUTPUT

```
canthi
canthi
canthi
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(t--)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
6
```

Test Case 2

TOKEN COUNT

```
210
```

Test Case 3

NLOC

```
36
```

Code

queen_game_wythoff.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;

    double phi = (1.0 + sqrt(5.0)) / 2.0;

    while(t--) {
        long long a, b;
        scanf("%lld %lld", &a, &b);
        long long x = (a < b) ? a : b;
        long long y = (a > b) ? a : b;
        long long k = y - x;

        long long cold_x = (long long)(k * phi);
        if (x == cold_x) {
            printf("sami\n");
        } else {
            printf("canthi\n");
        }
    }
    return 0;
}
```

Output

